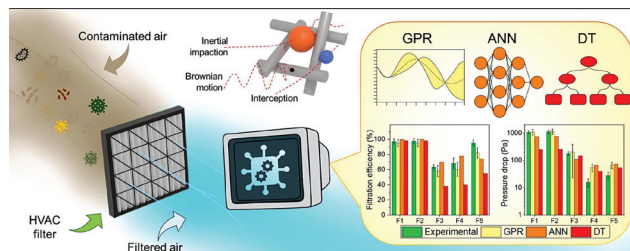


AI helps make better air filters for indoor

Researchers at IIT Delhi have developed an AI-based framework to design smarter air filters that balance clean air and energy efficiency. Using machine learning trained



IIT Delhi researcher-led team develops machine learning framework to design smarter HVAC filters to ensure healthier indoor air quality (Credit: <https://home.iitd.ac.in>)

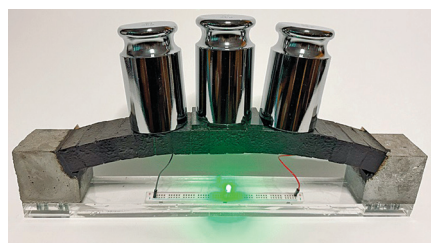
on global data, the model predicts how filter materials and structures affect airflow and particle capture before production. Validated with Elofic Industries, this innovation could help buildings, schools, hospitals, and offices, maintain healthier indoor air without higher power costs.

New polymers redefine high-frequency communications

Researchers at Waseda University, Japan, have unveiled a new class of polymers that could redefine high-frequency communications. Led by Professor Kenichi Oyaizu, the team developed poly (phenylene sulfide) derivatives that minimise signal loss by replacing oxygen with sulphur atoms. This breakthrough achieved an ultralow dissipation factor of 0.00087 at 10GHz, offering thermally stable, low-loss materials ideal for 6G networks, advanced radar, and high-speed computing.

Now walls, pavements, and bridges can be made to act like batteries

MIT (Cambridge, Massachusetts) engineers have developed electron-conducting carbon concrete (EC³), a new form of carbon-cement material that can store and release



Carbon-cement concrete that stores and releases electricity (Credit: <https://news.mit.edu>)

electricity like a supercapacitor. Made by blending cement, water, ultra-fine carbon black, and electrolytes, EC³ turns walls, pavements, and bridges into

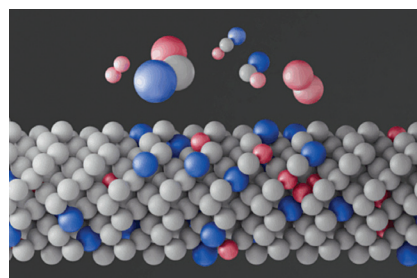
structural energy storage systems. With optimised electrolytes, it now needs just 5m³ to power a household, nearly nine times less than before, paving the way for smart, self-monitoring, energy-storing infrastructure.

Chip-scale circuits guide sound at gigahertz

Researchers from the University of Science and Technology of China, Penn State, and other institutes have developed compact phononic circuits that guide sound waves at 1.5GHz, enabling chip-scale devices for communications, sensing, and quantum technologies. These circuits manipulate phonons like electrons in electronics or photons in photonics, using microscopic waveguides and topological pathways to steer sound smoothly around defects. GHz operation ensures compatibility with microwave systems, while experiments confirm coherent, reconfigurable phonon transport. Scalable fabrication could integrate these circuits into hybrid systems, supporting advanced filters, precision sensors, and quantum components, paving the way for practical, robust chip-scale sound-based technologies.

New pathway to design semiconductors for next-gen devices

Researchers at Lawrence Berkeley National Laboratory and George Washington University have revealed that these atoms form localised patterns, or short-range order, which



An illustration of the semiconductor material investigated for this study, which is composed of germanium with small amounts of silicon and tin. The germanium atoms are depicted as gray spheres, the silicon as red and tin as blue (Credit: Minor et al/Berkeley Lab)

directly influence behaviour like the band gap. Engineers designing semiconductors for quantum computing, optoelectronics, and defence face a key challenge: trace elements in crystals can unpredictably alter electronic properties.

Using advanced 4D-STEM microscopy and machine-learning analysis, they mapped six atomic motifs, offering a new pathway to design semiconductors with precise electronic properties for next-generation devices.

New method boosts power efficiency and signal density in chips

As AI and HPC workloads outgrow traditional chip designs, Alphawave Semi has achieved a key milestone by completing the tape-out of its UCIE 3D IP using TSMC's advanced SoIC-X 3D packaging technology. This innovation tackles memory and bandwidth bottlenecks by enabling face-to-face die stacking that boosts power efficiency tenfold and signal density fivefold. The 5nm-3nm 3D integration marks a major leap toward high-performance, energy-efficient chiplet-based computing systems.